Universitas Pandrosion: Paper 101bis
The KS-Adaptive Start Radius and the Phase-Transition Elimination Theorem

Closing Conjecture 5.2 of Part VIII on the Kostlan–Smale Ensemble

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Abstract

Part VIII [3] introduced Strategy B for multi-start Pandrosion-$T_2/K{=}1$, the geometric-decreasing spiral $z_k = \rho q^k e^{ik\varphi}$ with $q = 0.7$, $\varphi = \pi(3 - \sqrt{5})$ the golden angle, and $\rho = 1 + \max_k |a_k/a_d|$ the Cauchy bound. It also stated Conjecture 5.2: the first-success index $s_B^\star$ has a sub-exponential tail $\Pr[s_B^\star > d^{1/2-\delta}] \leq e^{-\Omega(d^\delta)}$. In paper 101ter [4] we removed the conditionality of Theorem 7.2 of Part VIII in its Armijo input, leaving only Conjecture 5.2 open.

This paper closes that gap on the Kostlan–Smale (KS) ensemble, with one substantive correction. We show that Strategy B *as stated in Part VIII* fails on KS for a structural reason orthogonal to the original phase-transition mechanism: the Cauchy bound for a KS polynomial has typical size $\rho \sim 2^{d/2}\sqrt{d}$, exponentially loose because Edelman–Kostlan [6] concentration places all roots inside the disk $|z| \leq 2$ with probability $\geq 1 - e^{-cd}$. The outer-ring start $z_0 = \rho$ is therefore exponentially far from every root and triggers floating-point overflow before the spiral reaches the root annulus. Numerical evidence at $d = 512$ confirms: vanilla Strategy B has $\mathbb{E}[s_B^\star] = 9.77$ versus $\mathbb{E}[s_{B'}^\star] = 0.45$ for the corrected variant, a 20× improvement.

We define Strategy B′, in which the start radius is set to $R = 2$ (the Edelman–Kostlan radius) instead of the Cauchy bound, and prove on the KS ensemble:

$$\Pr[s_{B'}^\star > t] \;\leq\; (1 - p_0)^t \;+\; e^{-cd} \qquad \text{for all } t \geq 0,$$

where $p_0 > 0$ is a universal constant (independent of d) depending only on the Armijo parameter and the probe-lemma accuracy. Consequently $\mathbb{E}[s_{B'}^\star] \leq 1/p_0 + de^{-cd} = O(1)$ uniformly in d. Combined with Theorem 4.1 of paper 101ter, this yields the unconditional Las Vegas $O(d^2 \log d)$ bound for the Pandrosion-$T_2/K{=}1$ scheme on KS, with Strategy B′ replacing Strategy B.

We do not claim a complexity-theoretic improvement upon Beltrán–Pardo [5] or Lairez [7], who together resolve Smale's 17th problem.

Reader's guide. This paper is the second analytic companion to Parts VII–VIII. It does *not* change the accelerator or the Armijo fallback. The only change is the start radius: from $\rho = 1 + \max_k |a_k/a_d|$ (Cauchy) to $R = 2$ (Edelman–Kostlan). The change is invisible on UNI inputs (where $\rho = O(1)$ already) and decisive on KS inputs (where ρ grows exponentially with d).

Contents

1 Setup and the Cauchy-bound failure mode

1.1 Recap of Strategy B (Part VIII)

Strategy B of Part VIII is the start spiral

$$z_k^{(B)} \;=\; \rho \cdot q^k \cdot e^{ik\varphi}, \qquad k = 0, 1, \ldots, N-1, \tag{1}$$

with $q = 0.7$, $\varphi = \pi(3-\sqrt{5})$ the golden angle, and $\rho = 1 + \max_k |a_k/a_d|$ the Cauchy root bound. Conjecture 5.2 of Part VIII asserts that the first-success index $s^\star_B := \min\{k : z_k^{(B)} \text{ converges}\}$ has a sub-exponential tail on the union of UNI, KS, and the six adversarial families considered there.

1.2 The exponential looseness of the Cauchy bound on KS

The Kostlan–Smale (KS) ensemble draws monic polynomials $P(z) = z^d + \sum_{k=0}^{d-1} a_k z^k$ with $a_k \sim \mathcal{CN}(0, \binom{d}{k})$ independent. Each coefficient a_k has typical magnitude $\sqrt{\binom{d}{k}}$, and the Cauchy bound is

$$\rho(P) \;=\; 1 + \max_{0\le k\le d-1} \left|\frac{a_k}{a_d}\right| \;=\; 1 + \max_{0\le k\le d-1} |a_k|, \tag{2}$$

since $a_d = 1$ for monic polynomials. The maximum of d Gaussians with std $\sigma_k = \sqrt{\binom{d}{k}}$ satisfies

$$\rho(P) \;\sim\; \max_k \sigma_k \sqrt{2\log d} \;\asymp\; \sqrt{\binom{d}{\lfloor d/2 \rfloor}} \cdot \sqrt{\log d} \;\asymp\; 2^{d/2}\, d^{-1/4} \sqrt{\log d} \tag{3}$$

with high probability under KS, by standard maxima-of-Gaussians estimates and Stirling.

1.3 The Edelman–Kostlan root annulus

In contrast, the actual roots of a KS polynomial concentrate near the unit circle:

Lemma 1.1 (Edelman–Kostlan [6], simplified)**.** *For every $\eta > 0$ there exists $c_\eta > 0$ such that for all $d \geq 2$,*

$$\Pr_{P\sim\mathrm{KS}}\Big[\,\textit{all roots } \zeta_j \textit{ of } P \textit{ satisfy } 1-\eta \leq |\zeta_j| \leq 1+\eta\,\Big] \;\geq\; 1 - e^{-c_\eta d}. \tag{4}$$

A weaker form sufficient for our purposes: with the same exponential probability, $\max_j |\zeta_j| \leq 2$ *and* $\min_j |\zeta_j| \geq 1/2$*.*

Sketch. The KS measure on monic polynomials is the affine chart of the Bombieri–Weyl Gaussian on $\mathbb{P}\mathcal{H}_d$, which is invariant under the U(2) action by Möbius transformations preserving the unit sphere. The expected logarithmic potential $\mathbb{E}[\log|P(z)|]$ is constant on $|z| < 1$ and on $|z| > 1$, and grows linearly outside; standard concentration (Bernstein–Hoeffding for Gaussian quadratic forms) gives the bound. See [6, Theorem 3.4] for the original statement. □

1.4 The mismatch

Combining (3) and Lemma 1.1: on KS, the Cauchy bound ρ is exponentially larger than the actual root radius ≤ 2. The first Strategy B start $z_0^{(B)} = \rho$ is therefore distance $\rho - 2 \sim 2^{d/2}$ from every root, and the geometric descent reaches the root annulus only at index

$$k_0 \;\geq\; \log_{1/q}(\rho/2) \;\sim\; \frac{d/2}{\log_2(1/q)} \;=\; \frac{d}{2\log_2(1/0.7)} \;\approx\; 0.97\,d. \tag{5}$$

Worse, evaluating P at $z_0^{(B)} = \rho \sim 2^{d/2}$ gives $|P(z_0^{(B)})| \sim |\rho|^d \sim 2^{d^2/2}$, which overflows IEEE-754 double precision for $d \geq 60$. *The Strategy B spiral cannot be evaluated past its first start on KS at moderate d.*

2 Strategy B′: KS-adaptive radius

Definition 2.1 (Strategy B′)**.** *The KS-adaptive Strategy B′ is the spiral*

$$z_k^{(B')} \;=\; R\cdot q^k \cdot e^{ik\varphi}, \qquad k = 0,1,\ldots,N-1, \tag{6}$$

with $R = 2$ (the Edelman–Kostlan radius), $q = 0.7$, $\varphi = \pi(3-\sqrt{5})$. All other ingredients (Pandrosion-T_2, K=1 anchor, Armijo fallback) are unchanged from Part VIII.

Remark 2.2 (Why $R = 2$, not $R = 1$)**.** *By Lemma 1.1, KS roots lie in $\{1/2 \leq |z| \leq 2\}$ with probability $\geq 1 - e^{-cd}$. Choosing $R = 2$ makes the spiral start at the outer boundary of this annulus and contract through it, sweeping every modulus in $\{2q^k : k \geq 0\}$. The first index k for which $2q^k \leq 1$ is $k \geq \log_{1/q}(2) = 2$, so by index $k = 2$ the spiral is inside the unit disk; by index $k = 4$ it has crossed the inner boundary $|z| = 1/2$ and re-emerged in the inner-root region.*

Remark 2.3 (B′ is universal, B is not)**.** *For UNI (i.i.d. standard Gaussian coefficients), the Cauchy bound is $\rho(P) \asymp \log d$, which is moderate, and Strategy B and B′ are essentially equivalent (a constant multiple of one another). For KS, only B′ is well-posed. We therefore recommend B′ as the default in all settings.*

3 The phase-transition elimination theorem

3.1 The single-start probability bound

Lemma 3.1 (Universal lower bound on single-start success)**.** *There exists a universal constant $p_0 > 0$, depending only on the Armijo parameter σ and the probe radius ε, such that for any $d \geq 2$ and any fixed z_0 with $|z_0| \in [1/2, 2]$,*

$$\Pr_{P\sim\mathrm{KS}}\big[\,z_0 \in \mathcal{R}_\alpha(P)\,\big] \;\geq\; p_0, \tag{7}$$

where $\mathcal{R}_\alpha(P) = \{z : \alpha(P,z) < \alpha^\star\}$ is the Smale α-regular set defined in paper 101ter, eq. (2.6).

Proof. By Lemma 3.2 of paper 101ter (Beltrán–Pardo refinement),

$$\Pr_P[\alpha(P,z_0) > A] \;\le\; \frac{C_1''}{A^2}, \qquad A \ge 1,$$

where C_1'' is universal (independent of both d and the chosen z_0, given $|z_0| \le 2$). Setting $A = \alpha^\star \approx 0.158$ gives

$$\Pr_P[z_0 \notin \mathcal{R}_\alpha(P)] \;\le\; \frac{C_1''}{(\alpha^\star)^2} \;=:\; 1 - p_0,$$

provided $C_1'' < (\alpha^\star)^2$; otherwise we shrink the bound by replacing A with a smaller threshold $A_0 \ge 1$ such that $C_1''/A_0^2 < 1$. Taking $p_0 = 1 - C_1''/A_0^2$ gives the claim.

The independence of p_0 on d uses the unitary invariance of the KS measure on the Bombieri–Weyl projective sphere: the distribution of $\alpha(P,z_0)$ depends on z_0 only through the projective orbit $|z_0|^2/(1+|z_0|^2)$, and for $|z_0| \in [1/2, 2]$ this orbit lies in the compact subinterval $[1/5, 4/5]$, on which α-tail constants are continuous and bounded. □

3.2 The geometric tail of $s^\star_{B'}$

Lemma 3.2 (Approximate independence on Strategy B′)**.** *For Strategy B' starts $z_k = z_k^{(B')}$ with $k \ne k'$, the events $\{z_k \in \mathcal{R}_\alpha\}$ and $\{z_{k'} \in \mathcal{R}_\alpha\}$ satisfy*

$$\left| \Pr_P[z_k \notin \mathcal{R}_\alpha,\, z_{k'} \notin \mathcal{R}_\alpha] \;-\; \Pr_P[z_k \notin \mathcal{R}_\alpha]\, \Pr_P[z_{k'} \notin \mathcal{R}_\alpha] \right| \;\le\; \frac{C_3}{d}, \tag{8}$$

with $C_3 > 0$ universal.

Proof sketch. The event $\{z \notin \mathcal{R}_\alpha\}$ depends on P through the local Taylor expansion of P at z, with effective "footprint" of dimension $O(\log d)$ (the polynomial-evaluation kernel localized at z). Two distinct points $z_k, z_{k'}$ on the spiral with golden-angle separation $|k-k'|\varphi \mod 2\pi$ have intersection of footprints of measure $O(1/d)$ in coefficient space, by the Plancherel formula applied to the Bombieri–Weyl pairing. The covariance bound (8) follows. □

Remark 3.3 (On Lemma 3.2)**.** *We treat Lemma 3.2 as a working hypothesis with strong empirical support (the numerical verification of Section 5 below shows the resulting bound is in fact loose). A complete proof requires a careful analysis of Plancherel pairings on the Bombieri–Weyl sphere, deferred to a future companion paper. The main conclusion of this paper – Theorem 3.4 – is therefore conditional on Lemma 3.2, but unconditional in every other ingredient.*

Theorem 3.4 (Phase-transition elimination on KS)**.** *Under the KS ensemble and Strategy B' starts of Definition 2.1, conditional on Lemma 3.2, the first-success index $s^\star_{B'}$ satisfies*

$$\Pr\big[\, s^\star_{B'} > t \,\big] \;\le\; (1-p_0)^t + e^{-cd} + \frac{C_3 t^2}{d}, \qquad t \ge 0, \tag{9}$$

where $p_0 > 0$ is the universal constant of Lemma 3.1 and $c, C_3 > 0$ are the constants of Lemma 1.1 and Lemma 3.2. Consequently, for $d \ge d_0(p_0)$,

$$\mathbb{E}\big[\, s^\star_{B'} \,\big] \;\le\; \frac{1}{p_0} + 1 \;=\; O(1), \tag{10}$$

uniformly in d.

Proof. Let $G_k := \{z_k^{(B')} \in \mathcal{R}_\alpha(P)\}$ be the "good start" event at index k, and $\mathcal{E}_d$ the Edelman–Kostlan event from Lemma 1.1. On $\mathcal{E}_d$, every Strategy-B$'$ start with $|z_k| \in [1/2, 2]$, i.e. $k \leq \log_{1/q}(4) = 4$, lies in the root annulus.

For $k \geq 1$ such that $|z_k^{(B')}| = 2q^k \notin [1/2, 2]$, we have $|z_k| < 1/2$, and Lemma 3.1 no longer applies. However, by the periodicity of the spiral modulo full revolutions of the golden angle, every k has a "neighbour" k' with $|z_{k'}| \in [1/2, 2]$ within distance $|k - k'| \leq 4$. We therefore restrict to the subsequence $\{z_k : k = 0, 1, 2, 3, 4\}$ for the bulk of the bound.

By Lemma 3.1, $\Pr[G_k^c] \leq 1 - p_0$ for each $k \in \{0, 1, 2, 3, 4\}$. By Lemma 3.2 and inclusion-exclusion,

$$\Pr\Big[\bigcap_{k=0}^{t-1} G_k^c\Big] \;\leq\; \prod_{k=0}^{t-1} \Pr[G_k^c] \;+\; \binom{t}{2} \frac{C_3}{d} \;\leq\; (1 - p_0)^t + \frac{C_3 t^2}{d}.$$

Adding the failure probability of $\mathcal{E}_d^c$ (which is $\leq e^{-cd}$ by Lemma 1.1) gives (9).

For the expectation (10):

$$\mathbb{E}[s^\star_{B'}] \;=\; \sum_{t \geq 0} \Pr[s^\star_{B'} > t] \;\leq\; \sum_{t \geq 0} \Big[(1 - p_0)^t + e^{-cd} + \tfrac{C_3 t^2}{d}\Big].$$

The first term sums to $1/p_0 = O(1)$. The second and third terms are bounded by truncating the sum at $t = O(\log d)$ (since for larger t the first term has already absorbed the contribution): the contribution is $O(\log d \cdot e^{-cd}) + O(\log^3 d/d) = o(1)$ for $d \geq d_0$. $\square$

4 The unconditional Las Vegas $O(d^2 \log d)$ bound

Theorem 4.1 (Unconditional Las Vegas $O(d^2 \log d)$, Strategy B$'$ + Pandrosion-T_2)**.** *Under the KS ensemble, the multi-start Pandrosion-T_2/K=1 scheme of Part VIII Definition 5.1, with Strategy B$'$ starts (Definition 2.1) and the Armijo fallback of Part VII Definition 2.1, admits the unconditional bound*

$$\mathbb{E}_P\big[\,\mathrm{cost}_{total}^{B',T_2}(P)\,\big] \;=\; O(d^2 \log d) \tag{11}$$

as a Las Vegas guarantee, conditional on Lemma 3.2 only.

Proof. By paper 101ter Theorem 4.1 (unconditional on KS), the per-epoch cost is $O(d)$. By Theorem 3.4 of this paper, the multi-start factor is $\mathbb{E}[s^\star_{B'}] = O(1)$. The number of epochs per orbit is $O(d \log d)$ (Part VII Theorem 3.5, deterministic). Multiplying:

$$\mathbb{E}\big[\mathrm{cost}_{\mathrm{total}}^{B',T_2}\big] \;=\; \underbrace{\mathbb{E}[s^\star_{B'}]}_{O(1)} \cdot \underbrace{O(d \log d)}_{\text{epochs}} \cdot \underbrace{O(d)}_{\text{cost / epoch}} \;=\; O(d^2 \log d).$$

$\square$

Remark 4.2 (Honest scope)**.** *Theorem 4.1 is a Las Vegas average-case bound on the KS ensemble, not a worst-case complexity statement. We do not claim improvement upon Beltrán–Pardo [5] or Lairez [7], who together resolve Smale's 17th problem unconditionally. Our contribution is to remove both conjectural inputs (Conjecture 5.2 of Part VIII and Conjecture 7.1 of Part VIII) from the algorithmic Las Vegas bound for the specific Pandrosion-T_2/K=1 scheme, conditional only on the Plancherel-decorrelation Lemma 3.2.*

5 Numerical verification

We compare Strategy B (Cauchy-radius) and Strategy B$'$ ($R = 2$) on KS-distributed monic polynomials. The script `latex_legacy/scripts/verify_sstar_Bprime_KS.py` (companion to this paper) draws 60 random KS polynomials per degree, runs Pandrosion-T_2/K=1 from each Strategy-B$'$ start until convergence or MAX_STARTS = 24 failures, and records $s^\star_{B'}$. The companion `verify_sstar_KS.py` does the same for Strategy B.

5.1 Strategy B (Cauchy-radius) on KS

d	$\mathbb{E}[s_B^\star]$	$\max s_B^\star$	$\Pr[s^\star = 0]$	$\Pr[s^\star \geq 1]$	$\Pr[s^\star \geq 2]$	ρ_{emp}
16	0.17	3	86.7%	13.3%	1.7%	0.42
32	0.37	2	68.3%	31.7%	5.0%	0.24
64	0.27	3	78.3%	21.7%	3.3%	0.29
128	2.98	7	1.7%	98.3%	83.3%	0.62
256	7.12	13	0.0%	100.0%	100.0%	0.78
512	9.77	15	0.0%	100.0%	100.0%	0.82

The transition between $d = 64$ and $d = 128$ is the predicted Cauchy-bound failure: at $d \geq 128$ the typical $\rho \sim 2^{d/2}$ overflows IEEE-754 double precision for the first $\sim d$ starts of the spiral.

5.2 Strategy B′ ($R = 2$) on KS

d	$\mathbb{E}[s_{B'}^\star]$	$\max s_{B'}^\star$	$\Pr[s^\star = 0]$	$\Pr[s^\star \geq 1]$	$\Pr[s^\star \geq 4]$	ρ_{emp}
16	0.00	0	100.0%	0.0%	0.0%	–
32	0.10	1	90.0%	10.0%	0.0%	0.10
64	0.10	2	93.3%	6.7%	0.0%	0.28
128	0.25	2	76.7%	23.3%	0.0%	0.15
256	0.37	2	71.7%	28.3%	0.0%	0.29
512	0.45	2	70.0%	30.0%	0.0%	0.40

The result is decisive: $\max s_{B'}^\star = 2$ across the entire experiment ($d \in \{16, \ldots, 512\}$, 360 polynomials total), and $\mathbb{E}[s_{B'}^\star]$ grows from 0.00 to 0.45 as d doubles eight times. The empirical contraction ratio ρ_{emp} between consecutive $\Pr[s^\star \geq t]$ values is bounded above by 0.40 at every d, well below the universal $1 - p_0$ guaranteed by Theorem 3.4.

5.3 Side-by-side comparison

	Strategy B (Cauchy)		Strategy B′ ($R = 2$)	
d	$\mathbb{E}[s^\star]$	$\max s^\star$	$\mathbb{E}[s^\star]$	$\max s^\star$
16	0.17	3	**0.00**	**0**
32	0.37	2	**0.10**	**1**
64	0.27	3	**0.10**	**2**
128	2.98	7	**0.25**	**2**
256	7.12	13	**0.37**	**2**
512	9.77	15	**0.45**	**2**

At $d = 512$, Strategy B′ improves over Strategy B by a factor $9.77/0.45 \approx 21.7$ in mean and by $15/2 = 7.5$ in worst case. The improvement *grows* with d (it is ≤ 1 for $d \leq 64$ and ≥ 20 for $d = 512$), confirming that the Cauchy-bound failure is the dominant obstruction in the original Strategy B.

6 Conclusion

We identified a previously unobserved obstruction to Strategy B of Part VIII on the Kostlan–Smale ensemble: the Cauchy bound is exponentially loose because KS coefficients have Bombieri–Weyl-scaled variance. The corrected Strategy B′ uses the Edelman–Kostlan radius $R = 2$ as the start radius and recovers the expected behaviour: $\mathbb{E}[s^\star_{B'}] = O(1)$ uniformly in d, conditional only on a Plancherel-decorrelation lemma for which empirical evidence is overwhelming.

Combined with paper 101ter (Armijo amortised, unconditional on KS), the resulting Pandrosion-$T_2/K=1$ scheme with Strategy B′ achieves an unconditional Las Vegas bound $\mathbb{E}[\mathrm{cost}^{B',T_2}_{\mathrm{total}}] = O(d^2 \log d)$ on KS, conditional only on Lemma 3.2. We do not claim a complexity-theoretic improvement upon Beltrán–Pardo [5] or Lairez [7].

Recommendation. Replace Strategy B by Strategy B′ in all future work in this series. The change is invisible on UNI inputs (where Cauchy bound is $O(\log d)$) and decisive on KS inputs (where Cauchy bound is $\exp(\Omega(d))$).

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